

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

Ehco - Mail

Team Name: Team-Titans

Team Members:

- Puli Uma Vara Naga Durga Ujwal
 - Shaik Fareed Ahamad
 - Shaik Mobeen Arfath
 - Pokuru Venkata Phani Krishna Maheswar
 - Shaik Nazeeruddin
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Phase-1: Brainstorming & Ideation

Objective:

- AI Personalized Email Generator
- The primary purpose of an **AI Personalized Email Generator** is to automate and enhance the process of crafting tailored, relevant, and engaging email content for individual recipients.

Key Points:

1. **Problem Statement:** Many users struggle to write email and it's a time taking process and vocabulary mistakes
2. **Proposed Solution:** An **AI Personalized Email Generator** is a tool that uses artificial intelligence to create tailored and relevant email content for individual recipients.
3. **Target Users:** Businesses and Marketers, E-commerce and Retailers, Content Creators and Influencers, Event Planners and Organizations.
4. **Expected Outcome:** Automation of Personalized Email Creation, Enhanced Email Relevance.

Phase-2: Requirement Analysis

Objective:

Define technical and functional requirements for Eco-mail

Key Points:

1. Technical Requirements:

Programming Language: python.

Backend: Google Gemini Flash API.

Front end: Streamlit Web Frame Work.

Database: SQL

2. Functional Requirements:

Formating: Ability to create formatted Email using Gemini Flash API.

Data Integration: The system must be able to integrate with third-party data sources to enrich user profiles with real-time and historical data.

Contextual Content Creation: Based on user data, the system should be capable of generating contextually relevant content, such as product suggestions based on past textures or tailored recommendations based on user preferences.

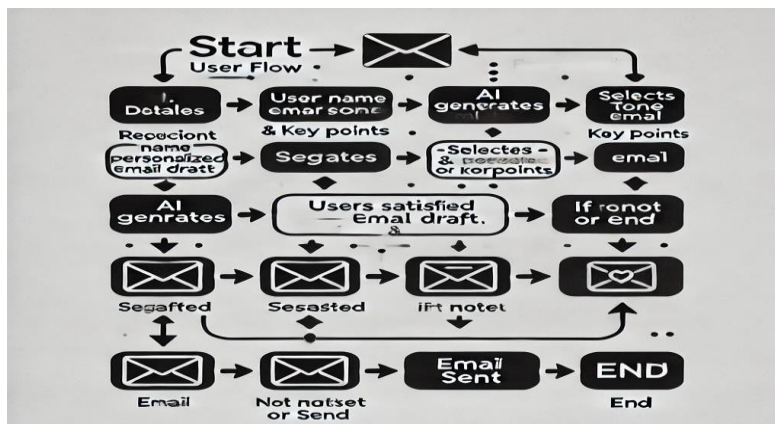
3. Constraints & Challenges:

Data Quality and Availability

Marketing providing best formatted structures.

Phase-3: Project Design

Objective:



Key Points:

1. System Architecture:

Frontend: Web/Mobile UI with input forms, email preview, and settings.
Backend: API layer, AI engine (NLP, personalization, formatting).
Database: Stores user preferences, templates, and email history.

2. User Flow:

User logs in.
Inputs recipient, subject, key points.
Selects tone & formality.
AI generates draft.
User reviews/edits.
Confirms & sends email.
AI learns from feedback.

3. UI/UX Considerations:

Minimalist UI: Clean, intuitive, responsive.
Guided Input: Pre-filled templates, dropdowns, real-time preview.
Smart Features: Auto-suggestions, one-click edits, AI feedback.

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down development tasks for efficient completion.

Sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Sprint 1	Environment Setup & API Integration	High	6 hours (Day 1)	End of Day 1	Puli Uma Vara Naga Durga Ujwal	Google API Key, Python, Streamlit setup	API connection established & working
Sprint 1	Frontend UI Development	Medium	2 hours (Day 1)	End of Day 1	Shaik Fareed Ahamad	API response format finalized	Basic UI with input fields
Sprint 2	EchoMail Search & Comparison	High	3 hours (Day 2)	Mid-Day 2	Pokuru Venkata Phani Krishna Maheswar	API response, UI elements ready	Search functionality with filters

Sprint 2	Error Handling & Debugging	🔴 High	1.5 hours (Day 2)	Mid-Day 2	Shaik Mobeen Arfath	API logs, UI inputs	Improved API stability
Sprint 3	Testing & UI Enhancements	🟡 Medium	1.5 hours (Day 2)	Mid-Day 2	Shaik Nazeeruddin	API response, UI layout completed	Responsive UI, better user experience
Sprint 3	Final Presentation & Deployment	🟢 Low	1 hour (Day 2)	End of Day 2	Puli Uma Vara Naga Durga Ujwal, Shaik Fareed Ahamad, Pokuru Venkata Phani Krishna Maheswar, Shaik Mobeen Arfath, Shaik Nazeeruddin	Working prototype	Demo-ready project

Sprint Planning with priorities :

Sprint 1 : Environment Setup & API Integration

- Priority: 🔴 High
- Duration: 6 hours (Day 1)
- Deadline: End of Day 1
- Assigned To: Puli Uma Vara Naga Durga Ujwal
- Dependencies: Google API Key, Python, Streamlit setup

- **Expected Outcome:** API connection established & working

☐ **Frontend UI Development**

- **Priority:** ☐ Medium
- **Duration:** 2 hours (Day 1)
- **Deadline:** End of Day 1
- **Assigned To:** Shaik Fareed Ahamad
- **Dependencies:** API response format finalized
- **Expected Outcome:** Basic UI with input fields

Sprint 2 : EchoMail Search & Comparison

- **Priority:** ☒ High
- **Duration:** 3 hours (Day 2)
- **Deadline:** Mid-Day 2
- **Assigned To:** Pokuru Venkata Phani Krishna Maheswar
- **Dependencies:** API response, UI elements ready
- **Expected Outcome:** Search functionality with filters

☐ **Error Handling & Debugging**

- **Priority:** ☒ High
- **Duration:** 1.5 hours (Day 2)
- **Deadline:** Mid-Day 2
- **Assigned To:** Shaik Mobeen Arfath
- **Dependencies:** API logs, UI inputs
- **Expected Outcome:** Improved API stability

Sprint 3 : Testing & UI Enhancements

- Priority: ☐ Medium
 - Duration: 1.5 hours (Day 2)
 - Deadline: Mid-Day 2
 - Assigned To: Shaik Nazeeruddin
 - Dependencies: API response, UI layout completed
 - Expected Outcome: Responsive UI, better user experience
 - ☐ Final Presentation & Deployment
 - Priority: ☐ Low
 - Duration: 1 hour (Day 2)
 - Deadline: End of Day 2
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Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. Technology Stack Used:

- **Frontend:** Streamlit
- **Backend:** Google Gemini Flash API
- **Programming Language:** Python

2. Development Process:

- Implement **API key authentication** and **Gemini API integration**.
- Develop **vehicle comparison and maintenance tips logic**.
- Optimize **search queries for performance and relevance**.

2. Challenges & Fixes:

- **Challenge:** Delayed API response times.
Fix: Implement **caching** to store frequently queried results.
- **Challenge:** Limited API calls per minute.
Fix: Optimize queries to fetch **only necessary data**.

Phase-6: Functional & Performance Testing

Objective:

Ensure the EchoMail App works as expected.

Test Category	Test Case	Expected Outcome	Priority	Status
Functional Testing	User Login & Authentication	Users should be able to log in securely	high	pending
Functional Testing	Input Data Validation	System should handle invalid inputs gracefully	high	pending
Functional Testing	AI Email Generation	AI should generate contextually relevant emails	high	pending
Performance Testing	Response Time for Email Generation	Email generation should not exceed 3 seconds	high	pending
Performance Testing	Load Testing	System should handle 1000+ concurrent users	high	pending
Performance Testing	Database Performance	AI should function under extreme load conditions	high	pending

Final Submission

1. **Project Report Based on the templates:** The AI-Personalized Email Generator is a tool designed to automate and personalize email drafting using AI. It enhances communication efficiency by generating context-aware, professional, and tailored emails based on user inputs.
2. **Demo Video (3-5 Minutes)**



hackthon.mp4

3. **GitHub/Code Repository Link:** <https://github.com/puliujwal143/Tech-Titans>
4. **Presentation:**



AI-Personalized-Email
-Generator-Crafting-tl

